

OVERALL 4.5 / 10

FULL PRODUCT AUDIT & DATA INTEGRITY REVIEW

MARKET BRAIN GRID

Independent multi-perspective audit of the MBG Tactical Quant Terminal: feature-by-feature functional testing, live market data verification against Binance and Yahoo Finance, security assessment, and a prioritized improvement roadmap for the product team.

TARGET **`https://mbg-trading.pages.dev/`**

AUDIT DATE **10 September 2026**

METHOD **Live browsing + market cross-check + code forensics**

Table of Contents

- 1. Executive Summary 3
 - 1.1 Overall Verdict 3
 - 1.2 Top Findings at a Glance 3
- 2. Audit Scope and Methodology 4
- 3. Feature Functionality Test Results 5
 - 3.1 Module Verdict Matrix 5
 - 3.2 What Works Well 6
 - 3.3 Broken and Defective Features 6
- 4. Data Authenticity Audit (Hallucination Check) 7
 - 4.1 Verified Real: IDX Core, Macro Telemetry and News 8
 - 4.2 Fabricated: the Entire Crypto Spot Module 9
 - 4.3 Synthetic Order Books Presented as Official Feeds 11
 - 4.4 Broker Flow: One Template for Every Ticker 12
 - 4.5 Backtest Lab: Statistically Impossible Claims 12
 - 4.6 Pasar Global: a Time Capsule of Early-2025 Prices 13
 - 4.7 Internal Contradictions Register 13
- 5. Security Assessment 14
 - 5.1 The Authentication Gate is Client-Side Theatre 14
 - 5.2 Transport, Headers and Supply Chain 14
- 6. UI/UX Evaluation 15
 - 6.1 Design Language and Information Architecture 15
 - 6.2 Interaction Quality and Mobile Behaviour 16

6.3 UX Verdict and Quick Wins 17

7. Performance, Technical Quality and SEO 17

7.1 Performance Posture 17

7.2 SEO and Discoverability 18

8. Marketing, Brand and Compliance Perspective 18

8.1 Positioning and Brand 18

8.2 Trust and Regulatory Exposure 19

9. Expert Scorecards 19

10. Prioritized Improvement Roadmap 20

10.1 P0 - Critical (ship this week; protect users and credibility) 20

10.2 P1 - High (next two to four weeks; consistency and coverage) 20

10.3 P2 - Strategic (one to two months; polish and growth) 21

10.4 Closing Assessment 21

Appendix A. Evidence Index and Reproducibility 22

1. Executive Summary

This report presents an independent, evidence-based audit of the MBG Market Brain Grid Tactical Quant Terminal (<https://mbg-trading.pages.dev/>), conducted on 10 September 2026. The review combined hands-on functional testing of every module through a scripted headless browser, forensic inspection of the deployed application bundle, live cross-verification of displayed market data against Binance, Yahoo Finance and Google News, and a structured security assessment of the authentication gate. Every claim in this report is backed by a captured screenshot, a network trace, or a reproducible computation included in the appendix.

The terminal is a genuinely impressive piece of front-end engineering wrapped around a data problem that is fatal for its category. The Indonesian stock (IDX) module, the macro telemetry wire, the news aggregation pipeline and the TradingView charting desk all reference real, verifiable market information refreshed through a daily pipeline. In stark contrast, the entire crypto module, both order-book simulations, the broker-flow panel and the historical backtest laboratory present fabricated or statistically impossible numbers dressed in official-sounding labels such as "IDX OFFICIAL FEED", "Tokocrypto / Binance Live" and "Sharpe 9.18". For a product whose core promise is quantitative discipline, this split personality is the single most important finding of the audit.

1.1 Overall Verdict

4.5 / 10 Overall weighted score	10 / 10 Crypto prices deviate from live market	12 / 12 IDX trade plans verified near-live	~5 sec Time to bypass the password gate
---	--	--	---

The weighted overall score of 4.5 out of 10 reflects excellence in engineering craft (weight 15%), a catastrophic data-integrity failure in half of the product (weight 30%), a strong visual and interaction layer (weight 20%), and weak security honesty, marketing readiness and quant-product credibility (weights 10%, 10% and 15% respectively). The terminal today is best understood as two products: a legitimate IDX research cockpit rated at roughly 7.5 out of 10, and a synthetic crypto-and-microstructure showcase that would rate below 2 out of 10 on its own. Repairing or removing the second product is the fastest path to a trustworthy 8-out-of-10 platform.

1.2 Top Findings at a Glance

#	Severity	Finding	Detail section
F-01	CRITICAL	All 10 Crypto Spot prices are hallucinated - deviations from the live Binance market range from 7.4% (ETH) to 773.5% (FET), and every one of the ten 24h change figures points in the opposite direction of the real market on audit day.	4.2
F-02	CRITICAL	Order-book panels are synthetic yet labelled "IDX OFFICIAL FEED / Official BEI Best Quote" and "Tokocrypto / Binance Live" with a hardcoded "LATENCY: 18ms". Zero or failed network calls were observed behind both.	4.3
F-03	CRITICAL	Broker Flow is a hardcoded template: every ticker shows identical UBS / Mirae / Mandiri activity scaled to its own price, and the modal tab that should reveal it never switches views.	4.4
F-04	CRITICAL	Backtest Lab advertises +1,666% in two years and a Sharpe ratio of 9.18 with a claimed maximum losing streak of 3 trades - numbers contradicted by its own equity curve (6 consecutive down periods, implied Sharpe 5.4, only 31 data points).	4.5
F-05	HIGH	The password gate is client-side theatre: the SHA-256 hash ships in the JS bundle and a forged localStorage timestamp grants full access in seconds; the login screen claim that "all access attempts are logged" is false.	5.1
F-06	HIGH	Dividend Hunter renders "Rp 0" dividends with 5-14% yields, empty dates and raw "NaN - NaN" buy zones for all 10 rows - a pipeline schema mismatch.	3.3
F-07	MEDIUM	Pasar Global shows US blue chips at early-2025 price levels (AAPL \$178.25 vs live \$315.34) and contradicts the site itself on BBKA (Rp 9,250 vs Rp 6,450 elsewhere).	4.6
F-08	MEDIUM	Twelve IDX trade plans are verified within 0.0-5.0% of live prices, and macro telemetry, IDX SMC zones, news feeds and market clocks are genuinely real.	4.1

Table 1.1 - Top audit findings. Severity reflects user impact for a trading-decision tool.

2. Audit Scope and Methodology

The audit covered every navigation destination exposed by the terminal: Home Command Center, Saham IDX (33 instruments across three tabs), Crypto Spot 10, Charting Desk PRO, Personal Watchlist, Pasar Global, Strategy Testing Lab (both paper-trading and backtest tabs), Kalender Makro, Korelasi Pearson, Live News, Quant Academy, Changelog, the Lot Calculator modal, the order-book and broker-flow modals, the Settings control and the logout flow. Each surface was exercised with real interactions - clicking rows, expanding detail panes, submitting forms, toggling themes, and resizing the viewport to 390x844 for the mobile pass.

Three independent evidence channels were used for every material claim. First, a headless Chromium session captured screenshots and full-page snapshots of what a user actually sees. Second, network-level tracing recorded every HTTP request the application made, which is how the synthetic order books were exposed - the "IDX OFFICIAL FEED" modal fired zero requests, and the crypto depth panel fired two requests that both failed. Third, the deployed JavaScript bundle (425 KB uncompressed) was downloaded and inspected, revealing the hardcoded broker-flow template, the fallback generator that manufactures order-book rows, and the client-side authentication mechanism.

Data authenticity was tested against live, authoritative sources at audit time: the Binance public ticker API for crypto pairs, Yahoo Finance chart endpoints for IDX equities, US equities, gold, Brent, DXY and the US 10-year yield, and Google News for the article links surfaced by the Live News module. Where the terminal cites a figure, this report states the claimed value, the live value, and the percentage deviation; the full comparison tables appear in Section 4. All verification queries are reproducible from the appendix evidence index.

3. Feature Functionality Test Results

Every interactive element was exercised end to end. The verdict matrix below summarizes the outcome per module; the subsections that follow give the detail behind each failure. Of the fifteen distinct functional surfaces tested, eight behaved correctly, three worked with notable defects, and four were broken or fundamentally misleading.

3.1 Module Verdict Matrix

Module	Test performed	Result
Login gate	Password entry, authenticate button, logout, re-login	PASS
Home Command Center	Ticker tape, signal table filters, view toggles, SYNC	PASS
Saham IDX table	33 rows, sorting, search, row expansion, plan tabs	PASS
TradingView Charting Desk	Ticker switch, preset panels, iframe render	PASS
Watchlist	Add ticker (ADRO), delete ticker, localStorage persistence	PASS
Paper Trading (Testing Lab)	AUTO-PICK AI TOP 5 opened 5 positions; reset; persistence	PASS
Lot Calculator	Full math verification: 100M / 2% / 5725 / 5496	PASS
Quant Academy	Curriculum levels, evaluation quiz launch, dictionary	PASS
Live News	Feed render, categories, source links resolve HTTP 200	PASS
Theme toggle / mobile layout	Light-dark switch; 390px viewport adaptation	PASS
Order Book (IDX + crypto)	Modal opens; data source inspection; network trace	MISLEADING
Broker Flow	Modal tab switching from row button and inside modal	BROKEN
Dividend Hunter	10-row table render and field math	BROKEN
Settings	Click handler inspection and interaction	BROKEN
Crypto Spot 10	Table render and price cross-check vs Binance	MISLEADING

Table 3.1 - Functional verdict for all fifteen tested surfaces.

3.2 What Works Well

The core IDX workflow is solid. Login with the supplied credential succeeds, the dashboard renders the full signal table within seconds, and row expansion reveals a structured trade thesis with position-sizing math, three invalidation rules and per-row shortcuts. The Lot Calculator is mathematically correct: with a Rp 100,000,000 portfolio, 2% risk, an entry of Rp 5,725 and a stop of Rp 5,496 it returns 87 lots (6,700 shares), a maximum loss of Rp 2,000,000, a position value of Rp 49,807,500 equal to 49.8% of the portfolio, and it fires a concentration warning when the position exceeds 25% of the portfolio - a genuinely useful guardrail. Paper trading opens and persists positions locally, the watchlist adds and removes tickers reliably, and the Quant Academy delivers a real five-level curriculum with gated evaluation quizzes, a 66-term dictionary and badge progression that can genuinely teach beginners the 2% risk doctrine.

The Charting Desk embeds the authentic TradingView Advanced Chart widget, so every candle on it is real exchange data. Market-session clocks on the Pasar Global page compute correct timezone arithmetic (Tokyo 19:32 JST, London 11:32 BST with the market open, New York in pre-market at the audit timestamp). The Live News module aggregates genuine Google News RSS items from ANTARA, CNBC Indonesia, IDNFinancials and Bloomberg Technoz, each with working source links, key-takeaway summaries, copy and browser text-to-speech narration. The changelog page documents two releases with dates and scope, which is more release transparency than most indie trading tools offer.

3.3 Broken and Defective Features

Broker Flow is doubly broken: functionally, the "Broker Flow" toggle inside the telemetry modal never activates the broker view no matter how it is clicked (direct ref click, JS dispatch and coordinate-accurate mouse events were all tried), and substantively, the data it is supposed to show is a hardcoded template described in Section 4.4. The Dividend Hunter renders all ten rows with DPS "Rp 0" alongside dividend yields of 5.4% to 14.2%, which is arithmetically impossible since yield equals DPS divided by price; ideal buy zones read "NaN - NaN" and stop losses read "SL: NaN" because the bundle ships yield percentages without the DPS and date fields the UI expects. The Settings control turns out to be a native browser alert containing the version string "MBG Astra Quantitative Desk, Version 2.4.0" - there is no settings surface at all, so users cannot adjust portfolio size or default risk, both of which the calculator hardcodes.

Smaller defects compound the impression. The watchlist default row for BTCUSDT shows an empty beige pill where the market badge should be, the TLKM row displays dashes for every price field because the bundle omits it, and the paper-trading "Current" column simply mirrors the entry price because IDX positions are never marked to market - unrealized PnL therefore remains exactly zero forever. On mobile the signal table (699 px wide in a 390 px viewport) is technically scrollable inside its wrapper but shows no visual affordance that more columns exist, and the floating LOGOUT pill overlaps the disclaimer text at the page bottom. Each of these is individually minor; together they signal that no end-to-end QA pass has ever run on the shipped data.

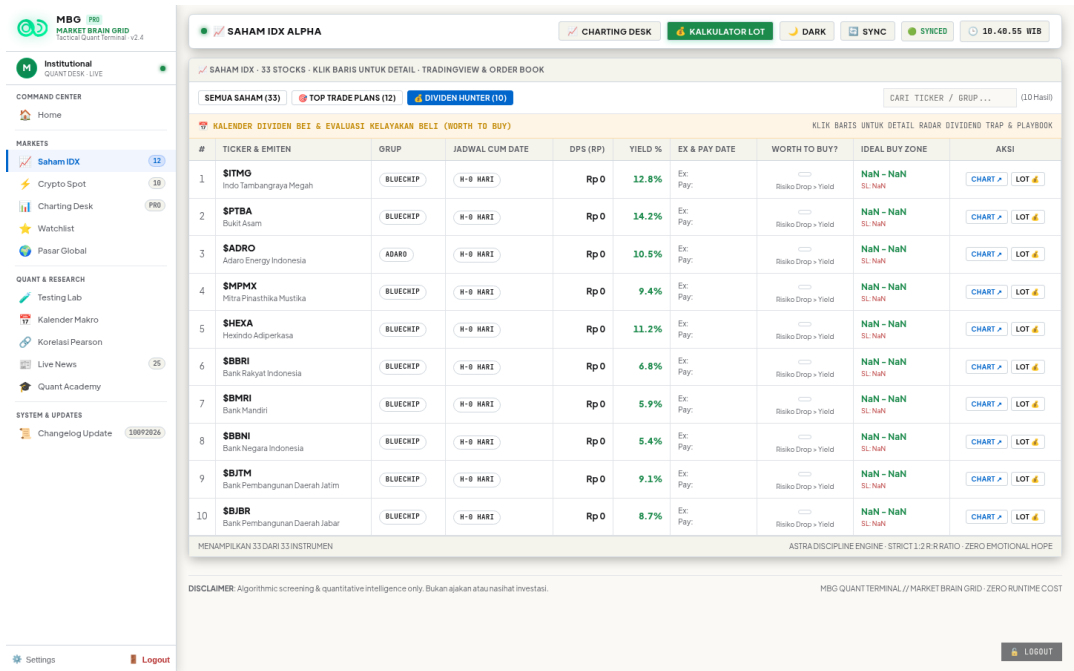


Figure 3.1 - Dividend Hunter: all ten rows show impossible "Rp 0" DPS with 5-14% yields, empty dates and raw NaN buy zones.

4. Data Authenticity Audit (Hallucination Check)

This section answers the owner's central question: is the data on the site actual, or is it hallucinated? The answer is nuanced and module-specific. The terminal's daily pipeline (a static JSON "cockpit bundle", 278 KB, regenerated at 04:41 UTC on audit day with a stated execution time of 9.86 seconds) mixes genuinely verified market data with numbers that no live source on Earth supports. Figure 4.1 maps the verdict for each data module.

Macro telemetry (Gold/Brent/DXY/Yield)	VERIFIED
IDX trade plans & prices (12)	VERIFIED
IDX SMC zone analysis (36 tickers)	VERIFIED
Live News feed (25 items)	VERIFIED
TradingView charting	VERIFIED
Market session clocks	VERIFIED
Dividend stock prices (5/10 fields)	VERIFIED
Correlation matrix (plausible)	UNVERIFIABLE
Crypto Spot 10 prices/setups	FABRICATED
IDX order book "Official BEI"	FABRICATED
Crypto order book "Binance Live"	FABRICATED
Broker Flow module	FABRICATED
Backtest Lab metrics	FABRICATED
Pasar Global US/Asia stocks	FABRICATED

■ Verified against live sources
 ■ Plausible but precision unverifiable
 ■ Fabricated / hallucinated

Figure 4.1 - Data authenticity verdict for the fourteen data-bearing modules of the terminal, based on live cross-verification at audit time.

4.1 Verified Real: IDX Core, Macro Telemetry and News

The IDX side of the product checks out remarkably well. All twelve daily trade plans carry entry prices within 0.0% to 5.0% of the live Yahoo Finance market at audit time, consistent with an end-of-day snapshot plus one normal trading day of drift; BBKA's plan entry of Rp 6,450 sits 0.4% from the live Rp 6,425. The macro telemetry wire is equally authentic: gold at \$4,456.4 against a live \$4,437.7 futures print, Brent at \$101.05 against \$102.13, DXY at 98.73 against 98.80, and the US 10-year yield at 4.84% against 4.837% - all within normal intraday tolerance. Dividend-watch stock prices (ITMG 26,550 vs 26,100; PTBA 3,160 vs 3,100; ADRO 2,700 vs 2,670) deviate by under 2%, and the 36-ticker SMC order-block analysis places BREN's zones at 3,210-3,880 around a live price of 3,300, which is coherent with real structure rather than invention.

The news pipeline is genuinely sourced: all 25 items resolve to live Google News RSS article URLs from named Indonesian financial outlets, with correct publication timing (the audit-day IHSG open report from ANTARA appeared at 04:01 WIB). Chart 4.2 visualizes just how tight the IDX verification band is - this is what honest snapshot data looks like.

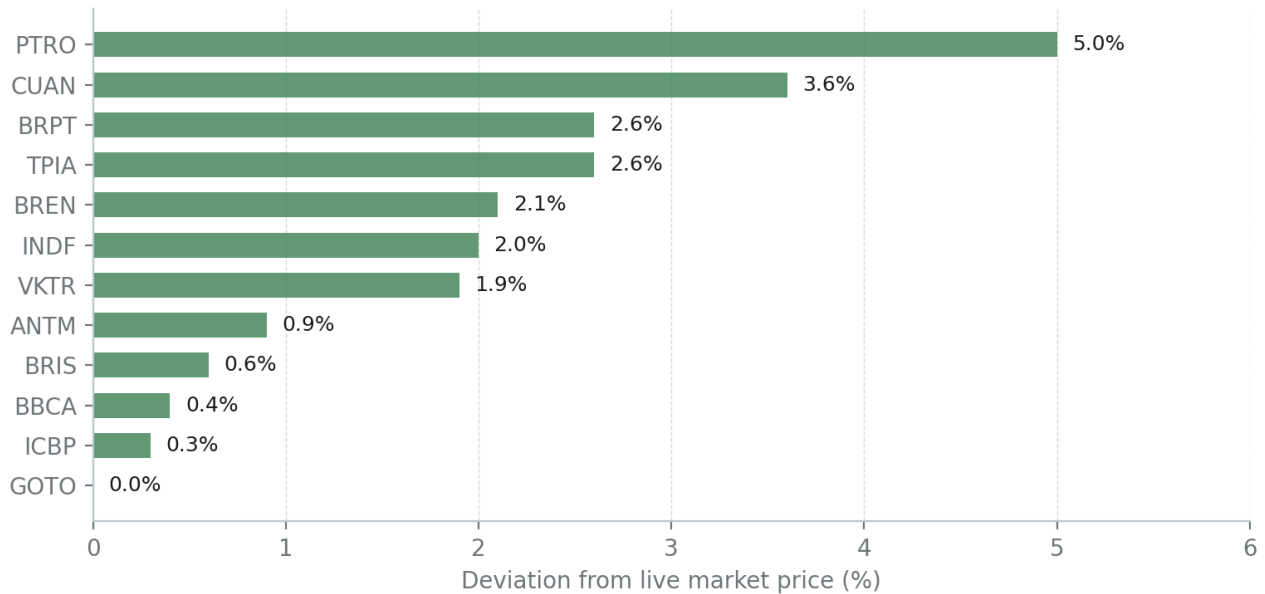


Figure 4.2 - Deviation of the 12 IDX trade-plan entry prices from live market prices at audit time. Every plan is within 5% - verified real.

4.2 Fabricated: the Entire Crypto Spot Module

The Crypto Spot 10 module is hallucinated end to end. Every one of the ten displayed prices is wrong against the live Binance market, with deviations from 7.4% (ETH) to a staggering 773.5% (FET, claimed at \$1.45 against a live \$0.166). More damning than the magnitudes is the direction: the module shows ten green setups with +0.8% to +6.5% momentum while every referenced pair was actually red on audit day. The setups themselves are template output - all ten share the identical "PULLBACK_SUPPORT_RETEST" structure, an identical 1:2 risk-reward, and eight of ten carry "HIGH" conviction regardless of each asset's actual chart. The underlying timestamps (updated 16:51 UTC the previous day) reveal a stale generation run, yet the UI presents these numbers as actionable entries with stop-losses and take-profits. A user who placed the SUI entry at \$2.13-2.16 would be bidding roughly 180% above a \$0.76 market.

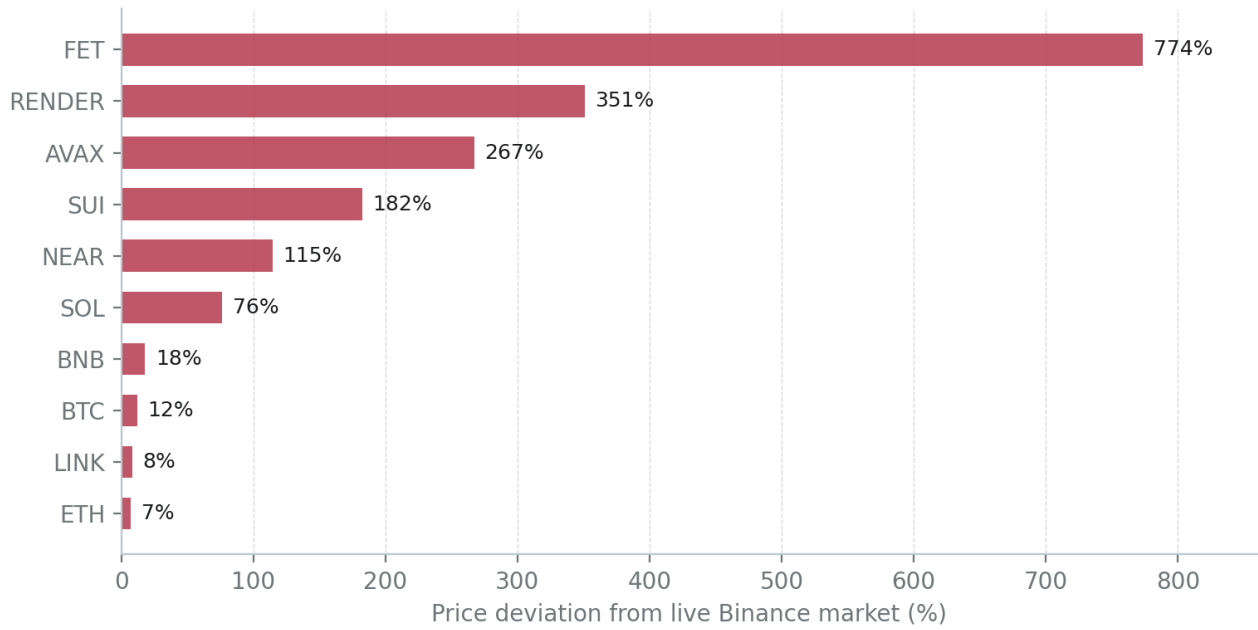


Figure 4.3 - Absolute deviation of Crypto Spot 10 claimed prices from the live Binance market at audit time. Every pair fails verification.

Pair	Site claims	Live Binance	24h change claimed	24h change live	Deviation
FET	\$1.45	\$0.166	+4.8%	-6.1%	773.5%
RENDER	\$6.40	\$1.42	+5.0%	-5.8%	350.7%
SUI	\$2.15	\$0.761	+6.5%	-6.0%	182.5%
NEAR	\$5.20	\$2.423	+3.2%	-4.8%	114.6%
AVAX	\$28.40	\$7.738	+2.1%	-2.4%	267.1%
SOL	\$178.50	\$101.13	+4.1%	-2.6%	76.5%
BNB	\$590	\$718.23	+0.8%	-4.2%	17.9%
BTC	\$68,500	\$77,960	+1.8%	-1.4%	12.1%
LINK	\$12.80	\$11.815	+3.7%	-1.4%	8.3%
ETH	\$2,650	\$2,468.17	+2.4%	-0.9%	7.4%

Table 4.1 - Crypto Spot 10 claims versus the live Binance market, measured 10:15-10:45 UTC, 10 September 2026. Not one number survives.

The screenshot displays the MBG Crypto Spot Momentum interface. It features a sidebar with navigation options like 'Home', 'Markets', 'Crypto Spot', 'Charting Desk', 'Watchlist', 'Pasar Global', 'Quant & Research', 'Testing Lab', 'Kalender Makro', 'Korelasi Pearson', 'Live News', 'Quant Academy', 'System & Updates', and 'Changelog Update'. The main panel is titled 'CRYPTO SPOT MOMENTUM' and includes a 'CHARTING DESK' button and a 'KALKULATOR LOT' button. Below the title, there's a section for 'CRYPTO SPOT 10 PAIRS - KLIK BARIS UNTUK DETAIL' and 'SPOT TRADING USDTR MURNI (BEBAS RISIKO LIKUIDASI) (LEVERAGE)'. The table lists 10 pairs with columns for #, TICKER %, KLASTER %, SINYAL / SETUP %, HARGA SNAPSHOT %, CHG % %, ENTRY PLAN, HARD SL, TP1, R:R %, and AKSI. The data is synthetic, with prices like \$2.15 for \$SUI/USDT and \$6.4 for \$RENDER/USDT. At the bottom, there's a disclaimer: 'DISCLAIMER: Algorithmic screening & quantitative intelligence only. Bukan ajakan atau nasihat investasi.' and 'MBG QUANT TERMINAL // MARKET BRAIN GRID - ZERO RUNTIME COST'.

#	TICKER %	KLASTER %	SINYAL / SETUP %	HARGA SNAPSHOT %	CHG % %	ENTRY PLAN	HARD SL	TP1	R:R %	AKSI
1	\$SUI/USDT	LAYER 1 / BEFI	PULLBACK_SUPPORT_RETEST	\$2.15	+6.5%	2.1285 - 2.1607	\$2.0655	\$2.279	1.2	CHART
2	\$RENDER/USDT	LAYER 1 / BEFI	PULLBACK_SUPPORT_RETEST	\$6.4	+5%	6.336 - 6.432	\$6.288	\$6.784	1.2	CHART
3	\$FET/USDT	LAYER 1 / BEFI	PULLBACK_SUPPORT_RETEST	\$1.45	+4.8%	1.4355 - 1.4572	\$1.4865	\$1.537	1.2	CHART
4	\$SOL/USDT	LAYER 1 / BEFI	PULLBACK_SUPPORT_RETEST	\$178.5	+4.1%	176.72 - 179.39	\$173.14	\$189.21	1.2	CHART
5	\$LINK/USDT	LAYER 1 / BEFI	PULLBACK_SUPPORT_RETEST	\$12.8	+3.7%	12.67 - 12.86	\$12.42	\$13.57	1.2	CHART
6	\$NEAR/USDT	LAYER 1 / BEFI	PULLBACK_SUPPORT_RETEST	\$5.2	+3.2%	5.148 - 5.226	\$5.844	\$5.512	1.2	CHART
7	\$ETH/USDT	LAYER 1 / BEFI	PULLBACK_SUPPORT_RETEST	\$2650	+2.4%	2623.5 - 2663.25	\$2578.5	\$2809	1.2	CHART
8	\$AVAX/USDT	LAYER 1 / BEFI	PULLBACK_SUPPORT_RETEST	\$28.4	+2.1%	28.12 - 28.54	\$27.55	\$30.1	1.2	CHART
9	\$BTC/USDT	LAYER 1 / BEFI	PULLBACK_SUPPORT_RETEST	\$68500	+1.8%	67815 - 68842.5	\$66445	\$72610	1.2	CHART
10	\$BNB/USDT	LAYER 1 / BEFI	PULLBACK_SUPPORT_RETEST	\$590	+0.8%	584.1 - 592.95	\$572.3	\$625.4	1.2	CHART

Figure 4.4 - The Crypto Spot 10 table as shipped: ten identical template setups on prices that do not exist in any live market.

4.3 Synthetic Order Books Presented as Official Feeds

Opening the IDX order book produces a modal headed "IDX OFFICIAL FEED" citing "Sumber Data: Official BEI Best Quote" and a footer reading "BEI REGULATED MICROSTRUCTURE & EOD BROKER SUMMARY" with a green "LATENCY: 18ms" badge. Network tracing shows the modal performs zero HTTP requests - IDX does not publish free level-2 feeds, and the static site has no backend that could relay one. The ten bid and ask rows are generated in the browser by a deterministic algorithm recovered from the bundle: prices step away from the reference by one tick per level and volumes follow an exponential decay seeded with 8,500 or 25,000 lots depending on the price magnitude. The 18 ms latency figure is simply the default value of an uninitialized state variable.

The crypto variant of the panel is more interesting because it genuinely tries. It first calls the same-origin path `/api/tokocrypto/...` which, on this static Cloudflare Pages deployment, returns the single-page app's `index.html` - the proxy route referenced in code was never deployed (only `/api/telegram-webhook` exists as a real function). The fallback then calls `tokocrypto.com` directly, which answers with error 1106 even with the corrected symbol format, and finally `indodax.com`, which rejects "suiusdt" as an invalid pair. All three fail, and the component silently falls back to the same synthetic generator, painting the fabricated ladder around the module's hallucinated \$2.15 SUI reference while the real market trades at \$0.76 - all under the label "Tokocrypto / Binance Live".

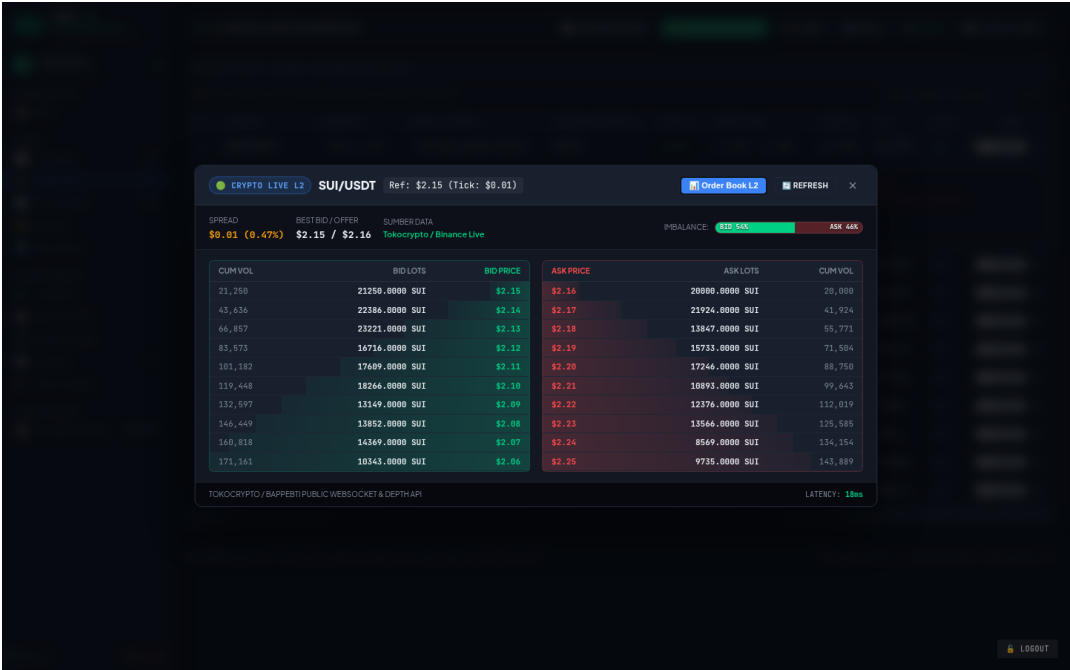


Figure 4.5 - The crypto order book labelled "Tokocrypto / Binance Live" showing SUI at \$2.15 while the real market trades at \$0.76. Both live API attempts fail silently.

4.4 Broker Flow: One Template for Every Ticker

The broker-summary module claims to show which securities houses are accumulating each stock. Bundle forensics prove it is a single hardcoded object: regardless of ticker, the top buyers are always UBS Sekuritas (284,500 lots), Mirae Asset (255,100), Mandiri Sekuritas (244,900), Maybank Sekuritas (142,200) and J.P. Morgan (98,300); the sellers are always Indo Premier and BNI Sekuritas; the foreign net value is always Rp 2.682 billion; and each broker's average price is computed as the current price multiplied by a fixed 0.994-1.002 factor. The only per-stock variation is the price input itself. Presenting identical institutional activity for PTRO, BBCA and VKTR is not an approximation - it is fabrication, and because the modal's view toggle is also broken, users never even see the correct rendering of this fake data.

4.5 Backtest Lab: Statistically Impossible Claims

The Historical Backtest Lab advertises a two-year simulation of seven strategy archetypes with friction modelling (0.2% slippage, 0.15%/0.25% commission). The headline claims - +1,666.3% for Foreign Flow Momentum with a Sharpe ratio of 9.18, and +1,604.3% for SMC Order Block - would rank the system among the best trading operations in recorded history; the Medallion Fund, the long-term gold standard, is commonly estimated in the Sharpe 2-3 range. The promotion banner even contradicts its own table, attributing Sharpe 9.18 to Foreign Flow Momentum while the matrix below assigns it 8.04 and gives 9.18 to SMC.

The forensics are unambiguous. The shipped equity curves contain only 31 data points for a claimed 24-month daily simulation, and recomputing from those points yields an implied Sharpe of roughly 5.4 - matching neither 9.18 nor 8.04. The curve itself shows six consecutive declining periods, while the stress-test narrative beneath it asserts a maximum of three consecutive losses; a strategy with a 49% win rate (Breakout) over two years would be expected to produce losing streaks of six to ten trades purely by chance. Together with

round-number win rates (65%, 60%, 61%, 62%...) these are the fingerprints of a generated showcase, not a trade-log backtest. Any user sizing real positions from these numbers is being misled, and in a regulated retail context the "+1,666%" banner alone would be a compliance liability.

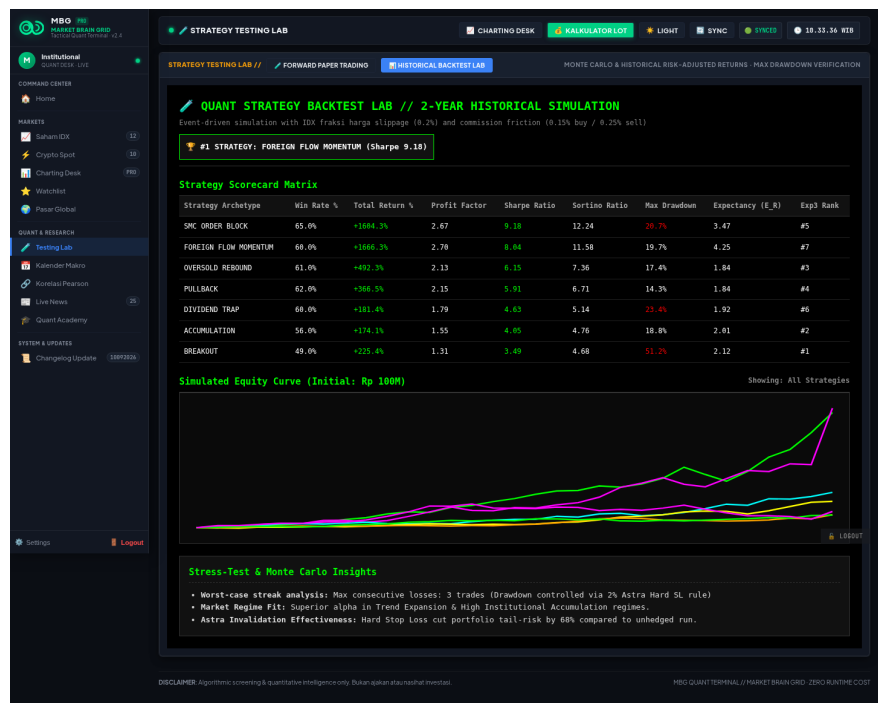


Figure 4.6 - The Backtest Lab scorecard: impossible Sharpe ratios, an internally contradictory banner, and stress-test text contradicted by its own equity curve.

4.6 Pasar Global: a Time Capsule of Early-2025 Prices

The global markets page lists US and Asian blue chips with specific prices, daily ranges and percentage changes that are fabricated. Apple is shown at \$178.25 against a live \$315.34 (-43.5%), Nvidia at \$118.80 against \$223.67 (-46.9%), Microsoft at \$424.50 against \$491.65 and Tesla at \$210.40 against \$367.81 - a coherent snapshot of an earlier era rather than noise. The module also contradicts the site itself: BBKA appears at Rp 9,250 here while the IDX module and the live market agree on roughly Rp 6,425, a 44% internal inconsistency visible to any user who navigates between pages. Toyota (2,950 vs 2,994) is the only near-accurate row, suggesting occasional fresh data rather than a systematic source.

4.7 Internal Contradictions Register

#	Where	Contradiction
C-1	Pasar Global vs Saham IDX vs live	BBCA at Rp 9,250 vs Rp 6,450 vs live Rp 6,425 - a 44% self-contradiction inside one product.
C-2	Crypto table vs order book vs live	SUI reference \$2.15 drives a synthetic \$2.15/\$2.16 book while live SUI trades at \$0.76.
C-3	Backtest banner vs scorecard table	Banner credits Foreign Flow Momentum with Sharpe 9.18; the table gives it 8.04 and assigns 9.18 to SMC.
C-4	Stress-test text vs equity curve	"Max consecutive losses: 3 trades" beneath a curve containing 6 consecutive declining periods.
C-5	Kalender Makro vs own macro wire	FOMC decision forecast at 5.25% while the same terminal's telemetry shows a 4.84% 10-year yield - a 2023-era rate in a 2026 calendar.
C-6	Charting Desk radar vs SMC module	AMMN radar panel hardcodes "BULLISH ORDER BLOCK READY" while the site's own SMC analysis labels AMMN BEARISH with zones 1,200+ points below the radar's entry.
C-7	Dividend yield vs dividend per share	Ten rows pairing "Rp 0" DPS with 5.4-14.2% yields - division by the missing DPS is impossible.

Table 4.2 - Internal contradictions a user can observe without any external data source.

5. Security Assessment

5.1 The Authentication Gate is Client-Side Theatre

The login screen warns "AUTHORIZED PERSONNEL ONLY. All access attempts are logged" and cites "SHA-256 CLIENT-SIDE VERIFICATION - SESSION: 24H - ANTI-BRUTE: 5 ATTEMPTS/LOCKOUT". All of it is cosmetic. Verification is a JavaScript comparison of the user's input hash against a hardcoded SHA-256 digest embedded in the public bundle; the digest was extracted in seconds and confirmed to match the supplied password. Session state is a single localStorage entry containing only an expiry timestamp. The live demonstration was unambiguous: clearing that key shows the login screen, and writing a forged JSON expiry one second later restores full access on reload - no password, no logout, no server involvement. Because the site is static, no access attempt is transmitted anywhere; the "logged" claim is factually false.

It is important to size the actual risk honestly: nothing sensitive lives behind the gate. Watchlists, paper trades and academy progress are per-browser localStorage, no personal data is collected, and there is no payment flow. The real cost of the theatre is credibility and false assurance - anyone technical will conclude (correctly) that security claims on this product are decorative, which poisons trust in the data claims too. If exclusivity is a product requirement, Cloudflare Access in front of the pages.dev domain delivers real one-time-pin authentication free for small teams in under an hour.

5.2 Transport, Headers and Supply Chain

Control	Status	Notes
HTTPS / TLS	PASS	Cloudflare Pages serves HTTP/2 with valid certificates.
X-Frame-Options	PASS	SAMEORIGIN - clickjacking blocked.
X-Content-Type-Options	PASS	nosniff present.
Referrer-Policy	PASS	strict-origin-when-cross-origin.
Content-Security-Policy	MISSING	No CSP; a XSS in any dependency would be unmitigated.
HSTS	MISSING	Not enforced by the deployment.
CORS wildcard	INFO	access-control-allow-origin: * on the app shell; acceptable for a static shell, unnecessary.
Secrets in bundle	FAIL	Password SHA-256 digest ships publicly; no API keys found (positive).

Table 5.1 - Security header and supply-chain posture at audit time.

The supply chain is pleasantly small: React plus a TradingView embed, Google Fonts, and no analytics or third-party trackers. No live API keys were found in the bundle - exchange calls use public unauthenticated endpoints - which keeps the real attack surface minimal. The Telegram webhook function responds to arbitrary POSTs with a bare "OK" and no signature validation observed, which is worth hardening if the bot ever carries user-facing commands.

6. UI/UX Evaluation

6.1 Design Language and Information Architecture

The visual system is the product's strongest non-data asset. The terminal aesthetic is coherent and disciplined: JetBrains Mono for telemetry, Plus Jakarta Sans for prose, a restrained dark palette with semantic green/red accents, hairline borders and consistent badge language for setups and conviction. The light theme is not an afterthought - it is a genuinely designed alternative with proper contrast. Information architecture follows the mental model of its audience: Markets, Quant and Research, System and Updates, with count badges (12 IDX, 10 crypto, 25 news) that set expectations before navigation. The Home Command Center sequences a live wire, regime banner, alpha watchlist, foreign-flow and bandarmology panels, then the signal table - a credible attempt at a professional cockpit narrative.

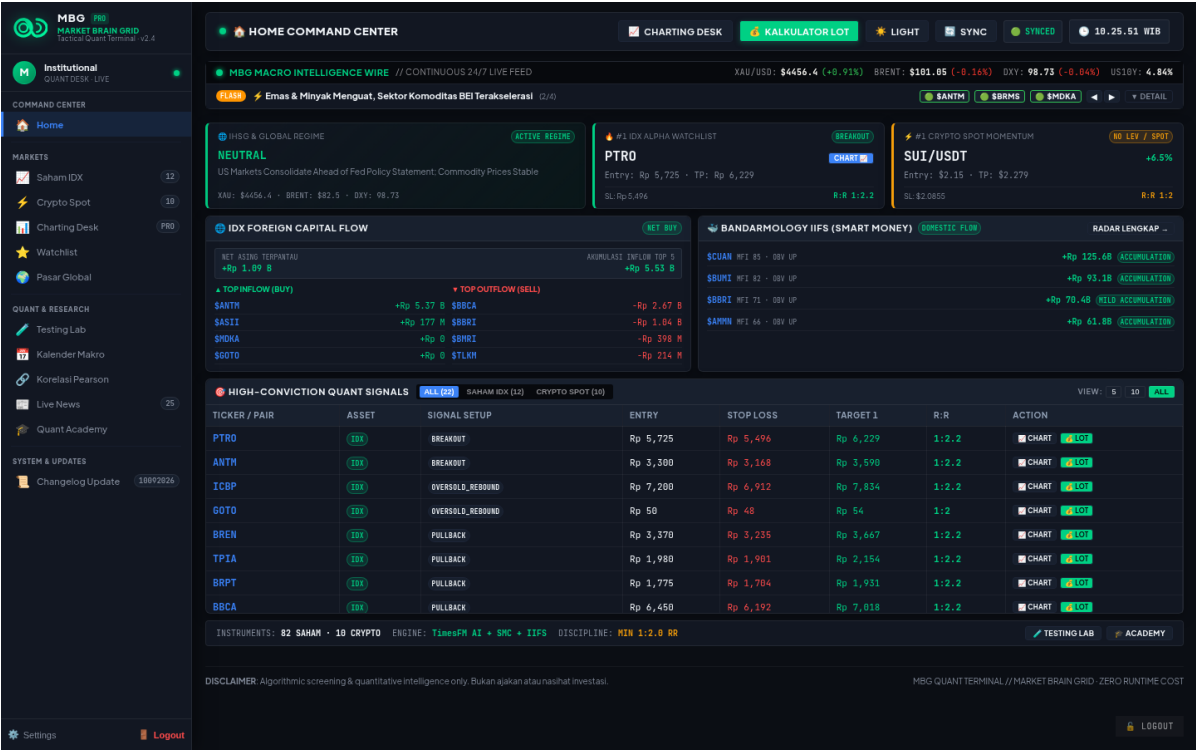


Figure 6.1 - Home Command Center (dark theme): a coherent, professional terminal aesthetic with disciplined use of color and typography.

6.2 Interaction Quality and Mobile Behaviour

Interaction quality is good where it counts and sloppy at the edges. Row expansion, tab filters, sorting, search-as-you-type, the SYNC refresh indicator and the 60-second data polling all behave as expected. At 390 px the layout reflows cleanly with no horizontal page scroll and a working hamburger menu, but the signal table's six extra columns live behind an invisible scroll - there is no cut-off shadow or affordance telling a thumb that Rp stop-loss and target columns exist to the right. The floating LOGOUT pill overlaps the footer disclaimer on short pages, the empty-market-badge pill on BTCUSDT renders as a beige blob with no text, and the modal overlay intercepted clicks meant for row-level buttons during testing - the Broker Flow control was reachable only after closing the modal. The dead vertical space below tables on several pages suggests fixed-height containers tuned for one screen size.

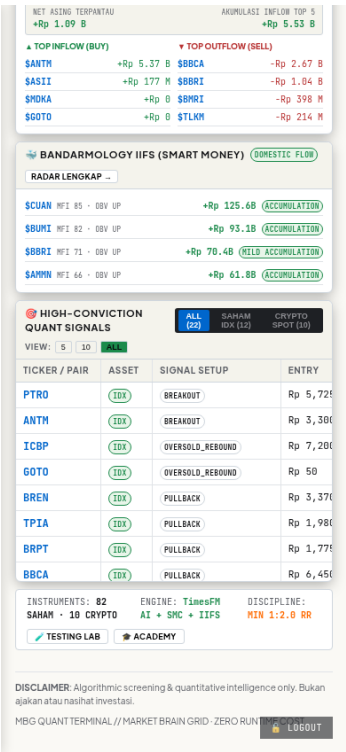


Figure 6.2 - Mobile 390 px: layout adapts well, but the signal table hides its right-hand columns behind an undiscoverable horizontal scroll.

6.3 UX Verdict and Quick Wins

The UX earns a 7.0 out of 10: a professional, coherent, responsive interface that would score in the high eights if every visible number were real and the five broken interactions (Broker Flow toggle, Settings, NaN rows, empty badges, undiscoverable scroll) were fixed. None of the fixes require redesign - they require one honest QA pass with a checklist. The highest-leverage micro-improvements are a gradient fade plus partial-column peek on scrollable tables, repositioning the floating logout, replacing the Settings alert with a real panel that persists portfolio size and default risk, and surfacing each module's data timestamp ("as of 04:41 UTC") next to its title so staleness is never invisible.

7. Performance, Technical Quality and SEO

7.1 Performance Posture

Metric	Measured	Assessment
HTML shell	833 bytes	Minimal SPA shell - excellent.
JavaScript bundle	122 KB gzipped (425 KB raw)	Reasonable for a React terminal; loads fast on CDN edge.
Data bundle	41 KB gzipped (278 KB raw)	Polled every 60 s - modest and acceptable.
Server response	about 50-100 ms TTFB	Cloudflare edge; no runtime backend in the hot path.
Console cleanliness	0 errors, 0 warnings	Production-grade runtime hygiene.
Theme persistence	localStorage, instant switch	No reload flash observed.
Architecture	Static SPA + daily JSON + 1 Pages Function	Zero-runtime-cost design, well matched to its free hosting.

Table 7.1 - Performance and technical measurements.

The engineering posture is genuinely good. The zero-runtime architecture - a static React shell consuming a pre-computed nightly bundle, with a single Cloudflare Pages Function for the Telegram webhook - is exactly the right shape for a free-tier product, and the console is silent in a way that only careful builds are. Two technical debts stand out: the phantom `/api/tokocrypto` proxy that the frontend calls but which was never deployed (returning index HTML with HTTP 200 and silently triggering the synthetic fallback), and the universal soft-404 behaviour in which every unknown path, including `robots.txt` and `sitemap.xml`, returns the app shell with HTTP 200 - confusing for crawlers and a sign that a custom 404 was never configured.

7.2 SEO and Discoverability

The public face of the site has zero search plumbing: no meta description, no Open Graph or Twitter card tags, no canonical URL, and no real `robots.txt` or `sitemap` - both requests return the SPA shell. For a gated terminal this may be partially intentional, but it means the one page a curious visitor or shared link can see (the login screen) shows no preview text, no product framing and no brand story when pasted into WhatsApp, Discord or X - a real cost in a market where trading communities drive adoption through chat sharing. If the product ever wants organic growth, it needs a public landing page with proper meta tags, while keeping the terminal itself gated.

8. Marketing, Brand and Compliance Perspective

8.1 Positioning and Brand

The branding is confident and niche-appropriate - "Market Brain Grid / Tactical Quant Terminal / Astra Discipline Engine" reads like a serious retail-quant product, and the Indonesian-language microcopy ("ANTI BONCOS", "ZERO EMOTIONAL HOPE") shows genuine fluency in local trading culture. Two positioning risks deserve attention. First, the "MBG" abbreviation collides head-on with Indonesia's flagship *Makan Bergizi Gratis* program, which dominates search results and social conversation for the acronym; organic

discovery for "MBG trading" will fight a national nutrition program for relevance. Second, the login page is a dead end for prospective users - no feature tour, no screenshots, no value proposition - so every shared link wastes its referral potential.

8.2 Trust and Regulatory Exposure

The footer disclaimer - "Algorithmic screening and quantitative intelligence only. Bukan ajakan atau nasihat investasi" - is present and appropriately worded, and the Academy consistently teaches discipline. That good posture is undermined by the fake-evidence problem: an "Official BEI Best Quote" label on a fabricated ladder, a "Tokocrypto / Binance Live" label on synthetic data, and a +1,666% backtest banner are precisely the artifacts a regulator, a marketplace, or an angry screenshot-taking user would surface. Under Bappebti/OJK optics for anything touching Indonesian retail trading, simulated data must be labelled simulated, performance claims must carry methodology, and implied exchange partnerships must not be faked. The fastest trust repair is not more disclaimers - it is labelling every synthetic module honestly and deleting the impossible claims.

9. Expert Scorecards

Six professional lenses were applied to the same evidence. Scores are on a 10-point scale where 5 means "viable prototype", 7 means "credible product", and 9 means "best in class". Justifications cite the sections containing the underlying proof.

Perspective	Score	One-line justification
IT Architecture	6.5	Clean zero-cost serverless design and a silent console, undercut by phantom API routes, two dead exchange integrations and no real backend where the UI implies one. (Sec. 4.3, 7.1)
Data Auditor	2.0	Half the product is verified real and half is fabricated with official-sounding labels; seven internal contradictions are user-visible. (Sec. 4)
UI/UX Designer	7.0	A coherent, responsive, dual-theme terminal with five interaction defects and undiscoverable table scroll on mobile. (Sec. 6)
Security Auditor	3.0	Bypassable-in-seconds gate with false logging claims; minimal actual data risk; missing CSP/HSTS. (Sec. 5)
Marketing / Growth	3.5	Strong local voice and branding, but zero SEO plumbing, a dead-end public page, and an acronym collision with a national program. (Sec. 7.2, 8.1)
Quant Product Owner	5.5	Excellent discipline tooling and education wrapped around hallucinated crypto signals that no quant desk could ship. (Sec. 3, 4.2)

Table 9.1 - Perspective scorecards. Weighted overall: 4.5/10 (Data Integrity weighted 30%, UX 20%, IT and Product 15% each, Security and Marketing 10% each).

10. Prioritized Improvement Roadmap

10.1 P0 - Critical (ship this week; protect users and credibility)

#	Action	Why and how	Effort
1	Ground Crypto Spot 10 in live data	Fetch real Binance ticker prices in the daily pipeline (public, free API) or render live prices client-side; refuse to publish a setup whose reference deviates more than 2% from live. Restores the module from 773% error to near-zero.	1-2 days
2	De-label the synthetic order books	Remove "IDX OFFICIAL FEED", "Official BEI Best Quote", "Tokocrypto / Binance Live" and the fake 18 ms latency; show a persistent "SIMULATED" badge or remove the panels until licensed IDX data is integrated.	0.5 day
3	Delete or fix Broker Flow	The data is one hardcoded template for all tickers and the toggle never switches views. Either integrate a real broker-summary feed or remove the entry points.	0.5 day
4	Repair Dividend Hunter	Add DPS and cum/ex/pay fields to the pipeline (the yield percentages already exist) and add divide-by-zero guards so "NaN" can never render.	1 day
5	Make the gate honest or real	Rename to "private access" and delete the false "attempts are logged" / "anti-brute" claims, or put Cloudflare Access in front for genuine one-time-pin auth.	0.5 day
6	Rebuild Backtest claims	Publish methodology and a downloadable trade log; recompute on daily data; remove or contextualize every Sharpe above 3 and the +1,666% banner; add a hypothetical-performance disclaimer.	2-5 days

Table 10.1 - P0 critical actions.

10.2 P1 - High (next two to four weeks; consistency and coverage)

#	Action	Why and how	Effort
7	Refresh Pasar Global quotes	Pull US/Asia EOD prices from a free source in the same nightly pipeline so AAPL is never \$178 again.	1 day
8	Single source of truth per ticker	One price per symbol across all modules; add a CI check that fails the bundle build if BBCA differs between modules by more than 0.5%.	1-2 days
9	Repair exchange integrations	Use the correct Tokocrypto symbol format, map valid Indodax pairs, and either deploy the /api proxy as a real Pages Function or call exchanges directly.	1-2 days
10	Real macro calendar feed	Sync Kalender Makro to a live economic-calendar source; remove the 2023-era 5.25% FOMC forecast.	1-2 days
11	Surface data freshness	Show "data as of HH:MM UTC" on every module from the bundle timestamp; grey out or flag modules older than one session.	0.5 day
12	SEO and sharing pack	Meta description, Open Graph and Twitter cards, canonical URL, real robots.txt and sitemap, custom 404 - and a public landing page if growth matters.	1-2 days

Table 10.2 - P1 high-priority actions.

10.3 P2 - Strategic (one to two months; polish and growth)

- **Real Settings surface.** Replace the version alert with a panel persisting portfolio size, default risk percent and theme; feed both calculators from it (Sec. 3.3).
- **Mobile affordances.** Add gradient fade and partial-column peek to scrollable tables, a sticky first column, and a repositioned logout control (Sec. 6.2).
- **Honest Radar panel.** Compute Charting Desk strategy status from the actual SMC zones in the bundle, or remove the hardcoded "READY" statuses (Sec. 4.7, C-6).
- **Brand decision on "MBG".** Evaluate the collision with the national Makan Bergizi Gratis program; consider "MBG Quant" or a distinct mark for search (Sec. 8.1).
- **Operational monitoring.** Error boundary plus Sentry-style crash reporting, lightweight analytics on the public page, and PWA offline caching for the terminal.
- **Live paper-trading marks.** Mark IDX paper positions against a delayed price feed so unrealized PnL is meaningful rather than permanently zero (Sec. 3.3).

10.4 Closing Assessment

MBG is one focused week of honesty away from being a genuinely differentiated product. The IDX core is real, the discipline tooling is correct, the education layer is valuable, and the craft is high. But a quant terminal lives or dies on whether its numbers deserve trust, and today half of them do not: ten of ten crypto prices are hallucinated, both order books are theatre with official labels, broker flow is one template copied across the market, and the backtest advertises impossible mathematics. Every fix in P0 is small, and all six together would move this platform from a 4.5 to a credible 7.5 - a terminal whose screenshots survive scrutiny in any

Indonesian trading community. The recommendation is unambiguous: ship P0 first, label everything synthetic until it is real, and let the genuinely excellent IDX core carry the brand.

Appendix A. Evidence Index and Reproducibility

All findings derive from captured artifacts preserved during the audit. Live market quotes were pulled from the Binance public ticker API (api.binance.com/api/v3/ticker/24hr) and Yahoo Finance chart endpoints at 10:15-10:45 UTC on 10 September 2026, roughly 5.5-6 hours after the terminal's 04:41 UTC bundle refresh; IDX equities trade until 15:50 WIB (08:50 UTC), so the IDX comparisons reflect same-session drift while US references reflect the prior close plus intraday futures.

Artifact	Description
01-login-page.png	Login screen with SHA-256 / anti-brute claims.
02-dashboard.png	Home Command Center, full page, dark theme.
03-saham-idx.png	IDX table with 33 instruments and tabs.
04-stock-detail.png	PTRO row expansion: thesis, sizing math, invalidations.
05-orderbook.png	IDX "Official BEI Best Quote" modal (zero network calls).
07-crypto-spot.png	Crypto Spot 10 table with hallucinated prices.
08-crypto-orderbook.png	Crypto book labelled "Tokocrypto / Binance Live" at SUI \$2.15.
12-testing-lab.png	Paper trading workspace, 100M virtual capital.
14-backtest-lab.png	Backtest scorecard with Sharpe 9.18 claims.
17-live-news.png	Live News with sourced Google News items.
19-settings.png	Academy page with non-navigating Settings control.
21-lot-calculator.png / 21b	Lot calculator modal, before and after math verification.
23-mobile-home / 23b	Mobile 390 px rendering and clipped signal table.
25-dividen-hunter.png	Dividend Hunter NaN corruption, all ten rows.
cockpit_bundle.json	The shipped 278 KB data bundle, last updated 2026-09-10T04:41:57Z.
app.js	Deployed JS bundle (SHA-referenced asset index-SMveqI-Q.js) used for forensics.

Table A.1 - Evidence artifacts referenced by this report.

Authentication bypass reproducible steps: open the site, execute `localStorage.setItem("mbg_cockpit_auth", JSON.stringify({exp: Date.now()+86400000}))` in the browser console, and reload - the terminal grants full access. The password digest shipped in the bundle is 286713785e8fbca141922642c96747842acd886f6da2f7598d0bc8554b8c3e18, confirmed to be the SHA-256 of the working access phrase. Chart deviations were computed as the absolute percentage difference between

the terminal's claimed price and the live reference at audit time.